

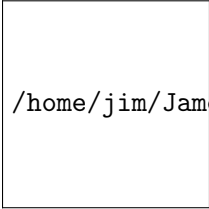
Section 1.2: Symbolic Logic

Finite Mathematics MTH 107 Fall 2014

Dr. James Ashe

Mars Hill University

Friday 12th September, 2014 and Monday 15th September, 2014



`/home/jim/JamesAshe/MHU/images/MarsHillU`

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- A triangle has three sides.

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False

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- A triangle has three sides. **T**
- Is Socrates a man?
- Socrates was a bad man.
- Go do some math.

STATEMENTS: True or

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STATEMENTS: True or False

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- $8 - 3 + 4$

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- A triangle has three sides. **T**

STATEMENTS: True or False

- Is Socrates a man?
- Socrates was a bad man.
- Go do some math.
- $8 - 3 + 4$
- This statement is false (*Liar's Paradox*).

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STATEMENTS: True or False

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NOT STATEMENTS

Compound Statements

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- Socrates was a man, and
Theano was a woman.

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- Socrates was a man, and Theano was a woman.
 - If a polygon has three sides then it is a triangle.

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- Socrates was a man, and Theano was a woman.
 - If a polygon has three sides then it is a triangle.
 - $8 - 3 + 4 = 9$ not 1.

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- If a polygon has three sides then it is a triangle.
- $8 - 3 + 4 = 9$ not 1.
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- If a polygon has three sides then go get some ice cream.

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 - If a polygon has three sides then go get some ice cream.
 - Truth or dare is a game.

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- If a polygon has three sides then it is a triangle.
- $8 - 3 + 4 = 9$ not 1.
- Is Socrates a man or woman?
- If a polygon has three sides then go get some ice cream.
- Truth or dare is a game.
- This statement is true, and this statement is false.

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COMPOUND STATEMENTS

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- Truth or dare is a game.
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NOT COMPOUND STATEMENTS

Your Turn: Statements

Problem 1.2.1. Decide whether the following is a statement. If it is a statement, decide whether or not it is compound

Nashville is the capital of Tennessee.

Your Turn: Statements

Problem 1.2.2. Decide whether the following is a statement. If it is a statement, decide whether or not it is compound

Got milk?

Your Turn: Statements

Problem 1.2.3. Decide whether the following is a statement. If it is a statement, decide whether or not it is compound

China does not have a population of more than 1 billion, and it is a country.

Your Turn: Statements

Problem 1.2.4. Decide whether the following is a statement. If it is a statement, decide whether or not it is compound

If I get an A then I will celebrate.

Negation

Negation

Negation

The *negation* of a true statement makes it false, and the *negation* of a false statement is true.

Statement

- Socrates is a man.

Negation

Negation

The *negation* of a true statement makes it false, and the *negation* of a false statement is true.

Statement

- Socrates is a man.

Negation of Statement

- Socrates is not a man.

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- Socrates is a man.
- Socrates is not a man.

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The *negation* of a true statement makes it false, and the *negation* of a false statement is true.

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- Socrates is a man.
- Socrates is not a man.

Negation of Statement

- Socrates is not a man.
- Socrates is a man.

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The *negation* of a true statement makes it false, and the *negation* of a false statement is true.

Statement

- Socrates is a man.
- Socrates is not a man.
- $8 - 3 + 4 = 9$

Negation of Statement

- Socrates is not a man.
- Socrates is a man.

Negation

Negation

The *negation* of a true statement makes it false, and the *negation* of a false statement is true.

Statement

- Socrates is a man.
- Socrates is not a man.
- $8 - 3 + 4 = 9$

Negation of Statement

- Socrates is not a man.
- Socrates is a man.
- $8 - 3 + 4 \neq 9$

Negation

Negation

The *negation* of a true statement makes it false, and the *negation* of a false statement is true.

Statement

- Socrates is a man.
- Socrates is not a man.
- $8 - 3 + 4 = 9$
- $3 < 4$, "3 is less than 4"

Negation of Statement

- Socrates is not a man.
- Socrates is a man.
- $8 - 3 + 4 \neq 9$

Negation

Negation

The *negation* of a true statement makes it false, and the *negation* of a false statement is true.

Statement

- Socrates is a man.
- Socrates is not a man.
- $8 - 3 + 4 = 9$
- $3 < 4$, "3 is less than 4"

Negation of Statement

- Socrates is not a man.
- Socrates is a man.
- $8 - 3 + 4 \neq 9$
- $3 \geq 4$, "3 is greater than or equal to 4"

Negation

Negation

The *negation* of a true statement makes it false, and the *negation* of a false statement is true.

Statement

- Socrates is a man.
- Socrates is not a man.
- $8 - 3 + 4 = 9$
- $3 < 4$, "3 is less than 4"
- All men are mortal.

Negation of Statement

- Socrates is not a man.
- Socrates is a man.
- $8 - 3 + 4 \neq 9$
- $3 \geq 4$, "3 is greater than or equal to 4"

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The *negation* of a true statement makes it false, and the *negation* of a false statement is true.

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- Socrates is a man.
- Socrates is not a man.
- $8 - 3 + 4 = 9$
- $3 < 4$, "3 is less than 4"
- All men are mortal.

Negation of Statement

- Socrates is not a man.
- Socrates is a man.
- $8 - 3 + 4 \neq 9$
- $3 \geq 4$, "3 is greater than or equal to 4"
- Some men are not mortal.

Negation

The *negation* of a true statement makes it false, and the *negation* of a false statement is true.

Statement

- Socrates is a man.
- Socrates is not a man.
- $8 - 3 + 4 = 9$
- $3 < 4$, "3 is less than 4"
- All men are mortal.
- No men are mortal.

Negation of Statement

- Socrates is not a man.
- Socrates is a man.
- $8 - 3 + 4 \neq 9$
- $3 \geq 4$, "3 is greater than or equal to 4"
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Negation

The *negation* of a true statement makes it false, and the *negation* of a false statement is true.

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- Socrates is a man.
- Socrates is not a man.
- $8 - 3 + 4 = 9$
- $3 < 4$, "3 is less than 4"
- All men are mortal.
- No men are mortal.

Negation of Statement

- Socrates is not a man.
- Socrates is a man.
- $8 - 3 + 4 \neq 9$
- $3 \geq 4$, "3 is greater than or equal to 4"
- Some men are not mortal.
- Some men are mortal.

Your Turn: Negation

Problem 1.2.5. Write a negation of each statement.

This is not the time to complain.

Your Turn: Negation

Problem 1.2.5. Write a negation of each statement.

This is not the time to complain.

Problem 1.2.6. Write a negation of each statement.

$$y \leq 5$$

Your Turn: Negation

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This is not the time to complain.

Problem 1.2.6. Write a negation of each statement.

$y \leq 5$

Problem 1.2.7. Write a negation of each statement

Some of the beverages contain caffeine.

Your Turn: Negation

Problem 1.2.5. Write a negation of each statement.

This is not the time to complain.

Problem 1.2.6. Write a negation of each statement.

$y \leq 5$

Problem 1.2.7. Write a negation of each statement

Some of the beverages contain caffeine.

Problem 1.2.8. Write a negation of each statement

None of the beverages contain caffeine.

Your Turn: Negation

Problem 1.2.5. Write a negation of each statement.

This is not the time to complain.

Problem 1.2.6. Write a negation of each statement.

$y \leq 5$

Problem 1.2.7. Write a negation of each statement

Some of the beverages contain caffeine.

Problem 1.2.8. Write a negation of each statement

None of the beverages contain caffeine.

Problem 1.2.9. Write a negation of each statement

All of the beverages contain caffeine.

Your Turn: Negation

Problem 1.2.5. Write a negation of each statement.

This is not the time to complain.

Problem 1.2.6. Write a negation of each statement.

$y \leq 5$

Problem 1.2.7. Write a negation of each statement

Some of the beverages contain caffeine.

Problem 1.2.8. Write a negation of each statement

None of the beverages contain caffeine.

Problem 1.2.9. Write a negation of each statement

All of the beverages contain caffeine.

Problem 1.2.10. Write a negation of each statement

Notation

	Symbol
statement	p, q, r , etc.
“and”	$\wedge$
“or”	$\vee$
“not”	$\sim$

Notation

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statement	p, q, r , etc.
“and”	$\wedge$
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“not”	$\sim$

Let m represent “Socrates is a man” and r represent “it’s going to rain”.

ex. *Socrates is a man and it’s going to rain.*

$$m \wedge r$$

ex. *Socrates is not a man or it’s going to rain.*

Notation

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Let m represent “Socrates is a man” and r represent “it’s going to rain”.

ex. *Socrates is a man and it’s going to rain.*

$$m \wedge r$$

ex. *Socrates is not a man or it’s going to rain.*

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Let m represent “Socrates is a man” and r represent “it’s going to rain”.

ex. *Socrates is a man and it’s going to rain.*

$$m \wedge r$$

ex. *Socrates is not a man or it’s going to rain.*

$$\sim m \vee r$$

ex. *It is not case that Socrates is a man or it’s going to rain.*

Notation

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statement	p, q, r , etc.
“and”	$\wedge$
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Let m represent “Socrates is a man” and r represent “it’s going to rain”.

ex. *Socrates is a man and it’s going to rain.*

$$m \wedge r$$

ex. *Socrates is not a man or it’s going to rain.*

$$\sim m \vee r$$

ex. *It is not case that Socrates is a man or it’s going to rain.*

$$\sim (m \vee r)$$

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$$\sim m \vee r$$

ex. *It is not case that Socrates is a man or it’s going to rain.*

$$\sim (m \vee r)$$

Equivalently, $\sim m \wedge \sim r$, *Socrates is not a man and it is not going to rain.*

Conditional Statements

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Conditional Statements

Notation

Symbol

statement p, q, r , etc.

" p and q " $p \wedge q$

" p or q " $p \vee q$

"not p " $\sim p$

"if p then q " $p \rightarrow q$

"If it is a mammal then it is warm blooded."

$p \equiv$ premise $q \equiv$ conclusion

Conditional Statements

Notation

Symbol

statement p, q, r , etc.

" p and q " $p \wedge q$

" p or q " $p \vee q$

"not p " $\sim p$

"if p then q " $p \rightarrow q$

p : It is a mammal

q : It is warm blooded.

ex. *If it is a mammal then it is warm blooded.*

$$p \rightarrow q$$

ex. *If it is not warmblooded then it is not a mammal.*

Conditional Statements

Notation

Symbol

statement	$p, q, r, \text{ etc.}$
" p and q "	$p \wedge q$
" p or q "	$p \vee q$
"not p "	$\sim p$
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p : It is a mammal

q : It is warm blooded.

ex. *If it is a mammal then it is warm blooded.*

$$p \rightarrow q$$

ex. *If it is not warmblooded then it is not a mammal.*

$$\sim q \rightarrow \sim p \text{ (contrapositive)}$$

Conditional Statements

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statement p, q, r , etc.

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ex. *All mammals are warm blooded.*

Conditional Statements

Notation

Symbol

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$$\sim q \rightarrow \sim p \text{ (contrapositive)}$$

ex. *All mammals are warm blooded.*

"If it is a mammal then it is warm blooded."

$p \equiv \text{premise}$ $q \equiv \text{conclusion}$

Conditional Statements

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Symbol

statement p, q, r , etc.

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p : It is a mammal

q : It is warm blooded.

ex. *If it is a mammal then it is warm blooded.*

$$p \rightarrow q$$

ex. *If it is not warmblooded then it is not a mammal.*

$$\sim q \rightarrow \sim p \text{ (contrapositive)}$$

ex. *All mammals are warm blooded.*

ex. *No snakes are warm blooded.*

Conditional Statements

Notation

Symbol

statement	p, q, r , etc.
" p and q "	$p \wedge q$
" p or q "	$p \vee q$
"not p "	$\sim p$
"if p then q "	$p \rightarrow q$

p : It is a mammal

q : It is warm blooded.

ex. *If it is a mammal then it is warm blooded.*

$$p \rightarrow q$$

ex. *If it is not warmblooded then it is not a mammal.*

$$\sim q \rightarrow \sim p \text{ (contrapositive)}$$

ex. *All mammals are warm blooded.*

ex. *No snakes are warm blooded.*

$$s \rightarrow \sim q$$

"If it is a snake then it is **not** warm blooded."

$s \equiv$ premise $\sim q \equiv$ conclusion

Necessary and Sufficient

Sufficient Condition

p : hypothesis

$\longrightarrow$

p implies q

Necessary Condition

q : conclusion

Necessary and Sufficient

Sufficient Condition

p : hypothesis

$\longrightarrow$

p implies q

Necessary Condition

q : conclusion

Necessary and Sufficient

Sufficient Condition

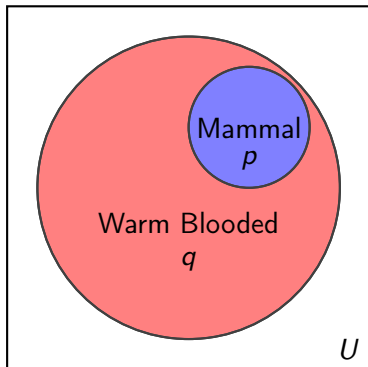
p : hypothesis

$\rightarrow$

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Necessary Condition

q : conclusion



Necessary and Sufficient

Sufficient Condition

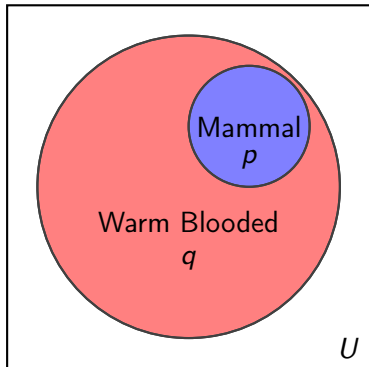
p : hypothesis

$\rightarrow$

p implies q

Necessary Condition

q : conclusion



p : It is a mammal

q : It is warm blooded.

ex. *Being a mammal is sufficient for being warm blooded.*

$$p \rightarrow q$$

ex. *Being warm blooded is necessary for being a mammal.*

Necessary and Sufficient

Necessary Condition

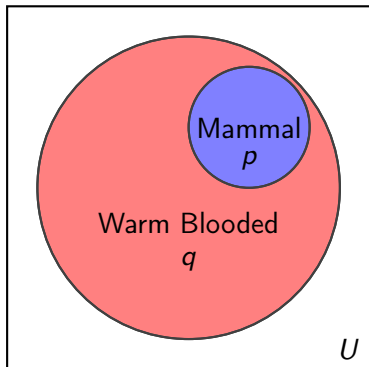
q : conclusion



q implies p

Sufficient Condition

p : hypothesis



p : It is a mammal

q : It is warm blooded.

ex. *Being a mammal is sufficient for being warm blooded.*

$$p \rightarrow q$$

ex. *Being warm blooded is necessary for being a mammal.*

$$q \leftarrow p$$

Necessary and Sufficient

Necessary Condition

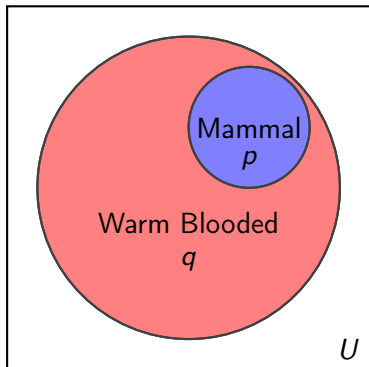
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Sufficient Condition

p : hypothesis



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ex. *Being warm blooded is necessary for being a mammal.*

$$q \leftarrow p$$

but we write $p \rightarrow q$.

Necessary and Sufficient

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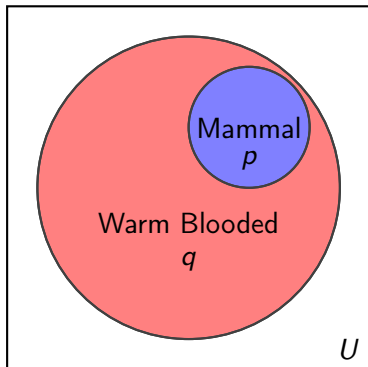
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Sufficient Condition

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ex. *Being a mammal is sufficient for being warm blooded.*

$$p \rightarrow q$$

ex. *Being warm blooded is necessary for being a mammal.*

$$q \leftarrow p$$

but we write $p \rightarrow q$.

- $q \nrightarrow p$: *Being warm blooded is not sufficient for being a mammal.*

Necessary and Sufficient

Necessary Condition

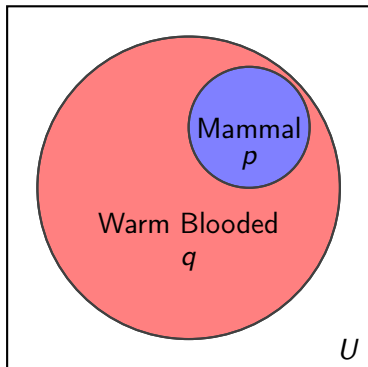
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Sufficient Condition

p : hypothesis



p : It is a mammal

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ex. *Being a mammal is sufficient for being warm blooded.*

$$p \rightarrow q$$

ex. *Being warm blooded is necessary for being a mammal.*

$$q \leftarrow p$$

but we write $p \rightarrow q$.

- $q \nrightarrow p$: *Being mammal is not necessary for being warm blooded.*

Problem 1.2.11. Using the symbolic representations

express the following compound statements in symbolic form.

Using Symbolic Representation

Problem 1.2.11. Using the symbolic representations

p : The food is spicy.

express the following compound statements in symbolic form.

- a. The food is aromatic and spicy.

Using Symbolic Representation

Problem 1.2.11. Using the symbolic representations

p : The food is spicy.

q : The food is aromatic.

express the following compound statements in symbolic form.

- a. The food is aromatic and spicy.
- b. If the food isn't spicy then it isn't aromatic.

Using Symbolic Representation

Problem 1.2.11. Using the symbolic representations

p : The food is spicy.

q : The food is aromatic.

express the following compound statements in symbolic form.

- a. The food is aromatic and spicy.
- b. If the food isn't spicy then it isn't aromatic.
- c. The food is spicy and it isn't aromatic.

Using Symbolic Representation

Problem 1.2.11. Using the symbolic representations

p : The food is spicy.

q : The food is aromatic.

express the following compound statements in symbolic form.

- a. The food is aromatic and spicy.
- b. If the food isn't spicy then it isn't aromatic.
- c. The food is spicy and it isn't aromatic.
- d. The food isn't spicy or aromatic.

Using Symbolic Representation

Problem 1.2.11. Using the symbolic representations

p : The food is spicy.

q : The food is aromatic.

express the following compound statements in symbolic form.

- a. The food is aromatic and spicy.
- b. If the food isn't spicy then it isn't aromatic.
- c. The food is spicy and it isn't aromatic.
- d. The food isn't spicy or aromatic.
- e. Being non-aromatic is sufficient for food to be non-spicy.

Using Symbolic Representation

Problem 1.2.12. Using the symbolic representations

p : The car costs \$70,000.

q : The car goes 140 mph.

r : the car is red.

express the following compound statements in symbolic form.

Using Symbolic Representation

Problem 1.2.12. Using the symbolic representations

p : The car costs \$70,000.

q : The car goes 140 mph.

r : the car is red.

express the following compound statements in symbolic form.

- a. All red cars go 140 mph.

Using Symbolic Representation

Problem 1.2.12. Using the symbolic representations

p : The car costs \$70,000.

q : The car goes 140 mph.

r : the car is red.

express the following compound statements in symbolic form.

- a. All red cars go 140 mph.
- b. The car is red, goes 140 mph, and does not cost \$70,000.

Using Symbolic Representation

Problem 1.2.12. Using the symbolic representations

p : The car costs \$70,000.

q : The car goes 140 mph.

r : the car is red.

express the following compound statements in symbolic form.

- All red cars go 140 mph.
- The car is red, goes 140 mph, and does not cost \$70,000.
- If the car does not cost \$70,000, it does not cost \$70,000.

Using Symbolic Representation

Problem 1.2.12. Using the symbolic representations

p : The car costs \$70,000.

q : The car goes 140 mph.

r : the car is red.

express the following compound statements in symbolic form.

- a. All red cars go 140 mph.
- b. The car is red, goes 140 mph, and does not cost \$70,000.
- c. If the car does not cost \$70,000, it does not cost \$70,000.
- d. The car is red, and it does not cost \$70,000 or go 140 mph.

Using Symbolic Representation

Problem 1.2.12. Using the symbolic representations

p : The car costs \$70,000.

q : The car goes 140 mph.

r : the car is red.

express the following compound statements in symbolic form.

- a. All red cars go 140 mph.
- b. The car is red, goes 140 mph, and does not cost \$70,000.
- c. If the car does not cost \$70,000, it does not cost \$70,000.
- d. The car is red, and it does not cost \$70,000 or go 140 mph.
- e. Being able to go 140 mph is sufficient for a car to cost \$70,000 or be red.

Using Symbolic Representation

Problem 1.2.12. Using the symbolic representations

p : The car costs \$70,000.

q : The car goes 140 mph.

r : the car is red.

express the following compound statements in symbolic form.

- a. All red cars go 140 mph.
- b. The car is red, goes 140 mph, and does not cost \$70,000.
- c. If the car does not cost \$70,000, it does not cost \$70,000.
- d. The car is red, and it does not cost \$70,000 or go 140 mph.
- e. Being able to go 140 mph is sufficient for a car to cost \$70,000 or be red.
- f. Not being red is necessary for a car to cost \$70,000 and not go 140 mph.